

Project Requirements - Sprint 6 Tasks

The following requirements are scoped for Sprint 6 implementation.

Task 1: Build RESTful Product Catalogue API

Implement Spring Boot REST endpoints for the product catalogue module.

Provide full CRUD for /products and /products/{id} with pagination support.

Add category filter, price sort, and keyword search query parameters.

Acceptance Criteria: All endpoints return correct HTTP codes. Pagination works.

Status: To Do

Task 2: Real-Time Inventory Tracking via WebSocket

Build an inventory service that pushes stock updates over WebSocket and STOMP.

Fire low-stock alerts when quantity falls below the configured minimum threshold.

Integrate with the order processing service to decrement stock on order placement.

Acceptance Criteria: Updates delivered in under 1 second. Alerts trigger correctly.

Status: In Progress

Task 3: Migrate Session Auth to OAuth 2.0 with OIDC

Replace the custom session-based authentication with OAuth 2.0 and OIDC.

Support Google and Microsoft as external identity providers.

Maintain backward compatibility with the existing username and password login.

Acceptance Criteria: OAuth login works. Social accounts linkable to user profile.

Status: Blocked

Task 4: Integrate Elasticsearch for Full-Text Product Search

Add Elasticsearch for full-text product search with autocomplete suggestions.

Support faceted filters by category, price range, brand, and average rating.

Sync data from PostgreSQL using Debezium Change Data Capture pipeline.

Acceptance Criteria: Search results returned in under 200ms. Filters accurate.

Status: To Do

Task 5: Build React Admin Analytics Dashboard

Create an admin dashboard in React using recharts for sales and revenue charts.

Support custom date range selection and CSV export of all filtered data.

Dashboard must load in under 2 seconds and be fully mobile responsive.

Acceptance Criteria: Charts render correctly. CSV exports all visible rows.

Status: To Do